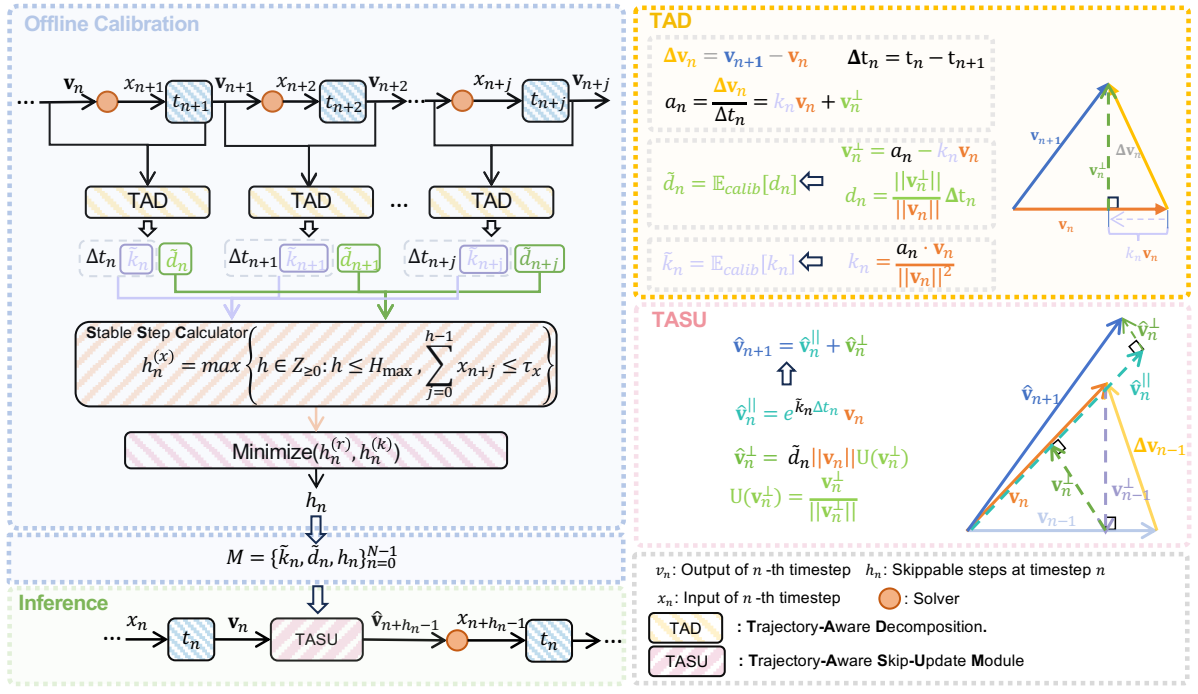

Offline Calibration
TAD
TAD
TAD
Stable Step Calculator
Minimize($h_n^{(r)}, h_n^{(k)}$)
h_n
$M = \{\tilde{k}_n, \tilde{d}_n, h_n\}_{n=0}^{N-1}$
Inference
TASU
TAD
$\Delta \mathbf{v}_n = \mathbf{v}_{n+1} - \mathbf{v}_n$
$\Delta t_n = t_n - t_{n+1}$
$a_n = \frac{\Delta \mathbf{v}_n}{\Delta t_n} = k_n \mathbf{v}_n + \mathbf{v}_n^{\perp}$
$\mathbf{v}_n^{\perp} = a_n - k_n \mathbf{v}_n$
$\tilde{d}_n = \mathbb{E}_{calib}[d_n]$
$d_n = \frac{||\mathbf{v}_n^{\perp}||}{||\mathbf{v}_n||} \Delta t_n$
$\tilde{k}_n = \mathbb{E}_{calib}[k_n]$
$k_n = \frac{a_n \cdot \mathbf{v}_n}{||\mathbf{v}_n||^2}$
TASU
$\hat{\mathbf{v}}_{n+1} = \hat{\mathbf{v}}_n^{||} + \hat{\mathbf{v}}_n^{\perp}$
$\hat{\mathbf{v}}_n^{||} = e^{\tilde{k}_n \Delta t_n} \mathbf{v}_n$
$\hat{\mathbf{v}}_n^{\perp} = \tilde{d}_n ||\mathbf{v}_n|| \mathrm{U}(\mathbf{v}_n^{\perp})$
$\mathrm{U}(\mathbf{v}_n^{\perp}) = \frac{\mathbf{v}_n^{\perp}}{||\mathbf{v}_n^{\perp}||}$
v_n: Output of n -th timestep
h_n: Skippable steps at timestep n
x_n: Input of n -th timestep
: Solver
TAD : Trajectory-Aware Decomposition.
TASU : Trajectory-Aware Skip-Update Module